

1.Import necessary packages

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

2.Load the file

```
In [4]: income_df = pd.read_csv(r'C:\Users\sande\OneDrive\Desktop\Naresh IT\data science an
income_df
```

Out[4]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_I
0	5000	8000	3	2000	
1	6000	7000	2	3000	
2	10000	4500	2	0	
3	10000	2000	1	0	
4	12500	12000	2	3000	
5	14000	8000	2	0	
6	15000	16000	3	35000	
7	18000	20000	5	8000	
8	19000	9000	2	0	
9	20000	9000	4	0	
10	20000	18000	4	8000	
11	22000	25000	6	12000	
12	23400	5000	3	0	
13	24000	10500	6	0	
14	24000	10000	4	0	
15	25000	12300	3	0	
16	25000	20000	3	3500	
17	25000	10000	6	0	
18	29000	6600	2	2000	
19	30000	13000	4	0	
20	30500	25000	5	5000	
21	32000	15000	4	0	
22	34000	19000	6	0	
23	34000	25000	3	4000	
24	35000	12000	3	0	
25	35000	25000	4	0	
26	39000	8000	4	0	
27	40000	10000	4	0	
28	42000	15000	4	0	
29	43000	12000	4	0	

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_H
30	45000	25000	6	0	
31	45000	40000	6	3500	
32	45000	10000	2	1000	
33	45000	22000	4	2500	
34	46000	25000	5	3500	
35	47000	15000	7	0	
36	50000	20000	4	0	
37	50500	20000	3	0	
38	55000	45000	6	12000	
39	60000	10000	3	0	
40	60000	50000	6	10000	
41	65000	20000	4	5000	
42	70000	9000	2	0	
43	80000	20000	4	0	
44	85000	25000	5	0	
45	90000	48000	7	0	
46	98000	25000	5	0	
47	100000	30000	6	0	
48	100000	50000	4	20000	
49	100000	40000	6	10000	

In [5]: `income_df.head()`

Out[5]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_H
0	5000	8000	3	2000	
1	6000	7000	2	3000	
2	10000	4500	2	0	
3	10000	2000	1	0	
4	12500	12000	2	3000	



3. Analyze the data

```
In [6]: income_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income                       50 non-null     int64
1   Mthly_HH_Expense                      50 non-null     int64
2   No_of_Fly_Members                    50 non-null     int64
3   Emi_or_Rent_Amt                      50 non-null     int64
4   Annual_HH_Income                     50 non-null     int64
5   Highest_Qualified_Member              50 non-null     object
6   No_of_Earning_Members                 50 non-null     int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB
```

```
In [7]: income_df.shape
```

```
Out[7]: (50, 7)
```

```
In [8]: income_df.describe().T
```

```
Out[8]:
```

	count	mean	std	min	25%	50%	
Mthly_HH_Income	50.0	41558.00	26097.908979	5000.0	23550.0	35000.0	50
Mthly_HH_Expense	50.0	18818.00	12090.216824	2000.0	10000.0	15500.0	25
No_of_Fly_Members	50.0	4.06	1.517382	1.0	3.0	4.0	
Emi_or_Rent_Amt	50.0	3060.00	6241.434948	0.0	0.0	0.0	3
Annual_HH_Income	50.0	490019.04	320135.792123	64200.0	258750.0	447420.0	594
No_of_Earning_Members	50.0	1.46	0.734291	1.0	1.0	1.0	



```
In [9]: income_df.isna().any()
```

```
Out[9]: Mthly_HH_Income      False
Mthly_HH_Expense         False
No_of_Fly_Members        False
Emi_or_Rent_Amt           False
Annual_HH_Income          False
Highest_Qualified_Member  False
No_of_Earning_Members     False
dtype: bool
```

4.What is the Mean Expense of a Household?

```
In [10]: income_df["Mthly_HH_Expense"].mean()
```

```
Out[10]: np.float64(18818.0)
```

5.What is the Median Household Expense?

```
In [11]: income_df["Mthly_HH_Expense"].median()
```

```
Out[11]: 15500.0
```

6.What is the Monthly Expense for most of the Households?

```
In [12]: mth_exp_tmp = pd.crosstab(index=income_df["Mthly_HH_Expense"], columns='count')
mth_exp_tmp.reset_index(inplace=True)
mth_exp_tmp[mth_exp_tmp['count'] == income_df.Mthly_HH_Expense.value_counts().max()]
```

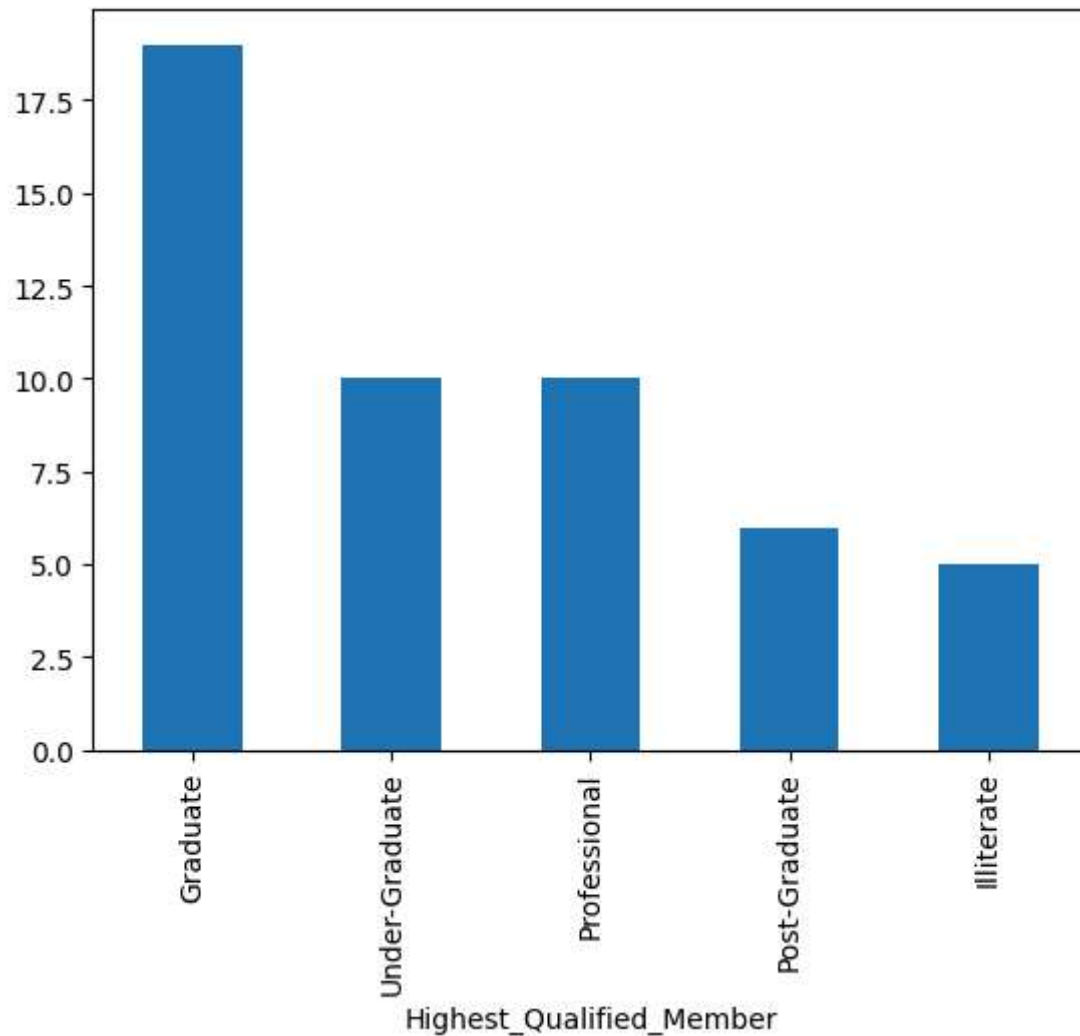
```
Out[12]:
```

col_0	Mthly_HH_Expense	count
18	25000	8

7.Plot the Histogram to count the Highest qualified member

```
In [13]: income_df["Highest_Qualified_Member"].value_counts().plot(kind="bar")
```

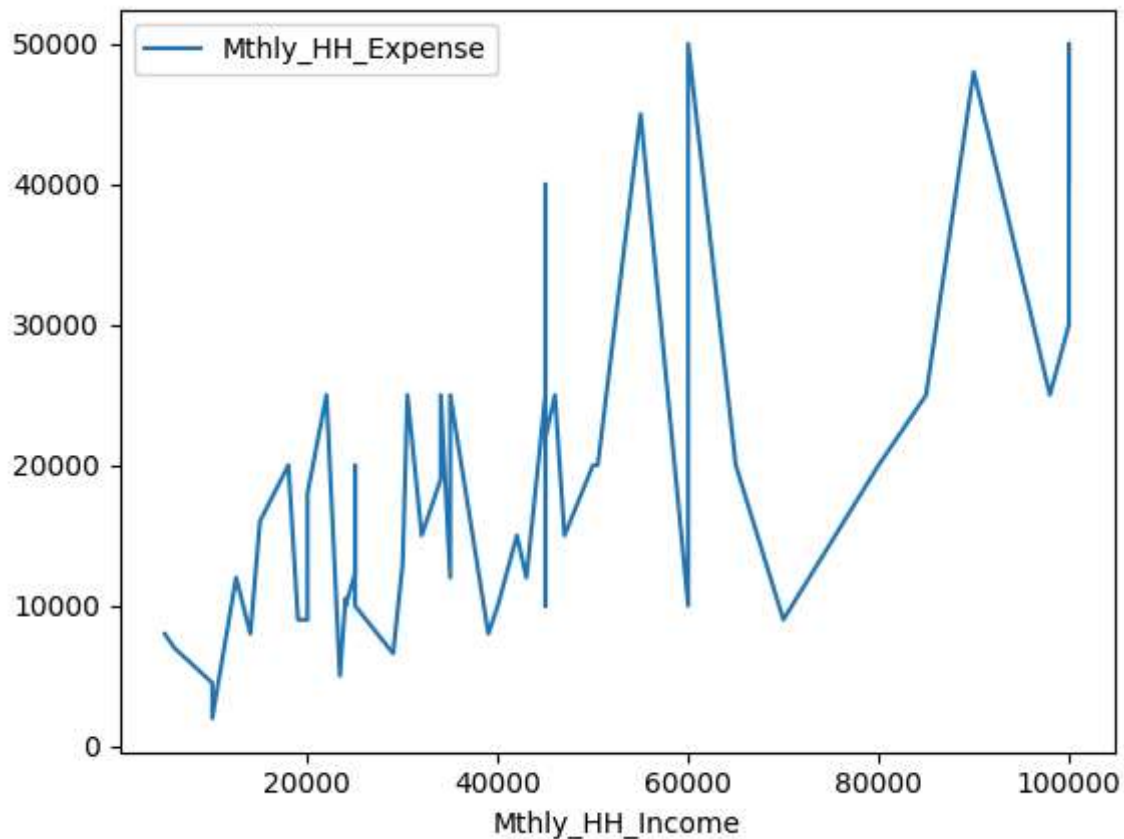
```
Out[13]: <Axes: xlabel='Highest_Qualified_Member'>
```



8.Calculate IQR(difference between 75% and 25% quartile)

```
In [14]: income_df.plot(x="Mthly_HH_Income", y="Mthly_HH_Expense")
IQR=income_df["Mthly_HH_Expense"].quantile(0.75)-income_df["Mthly_HH_Expense"].quan
IQR
```

```
Out[14]: np.float64(15000.0)
```



9. Calculate Standard Deviation for first 4 columns.

```
In [22]: pd.DataFrame(income_df.iloc[:,0:5].std().to_frame()).T
```

```
Out[22]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HI
0	26097.908979	12090.216824	1.517382	6241.434948	3201:



10. Calculate Variance for first 3 columns.

```
In [30]: pd.DataFrame(income_df.iloc[:,0:4].var().to_frame()).T
```

```
Out[30]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt
0	6.811009e+08	1.461733e+08	2.302449	3.895551e+07

11. Calculate the count of Highest qualified member.

```
In [31]: income_df["Highest_Qualified_Member"].value_counts().to_frame().T
```

```
Out[31]:
```

	Highest_Qualified_Member	Graduate	Under-Graduate	Professional	Post-Graduate	Illiterate
	count	19	10	10	6	5

12. Plot the Histogram to count the No_of_Earning_Members

```
In [32]: income_df["No_of_Earning_Members"].value_counts().plot(kind="bar")
```

```
Out[32]: <Axes: xlabel='No_of_Earning_Members'>
```

